

SAMUEL LEVENTHAL

CONTACT INFORMATION

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ABOUT

Ph.D. in Computer Science from the University of Utah (Scientific Computing and Imaging Institute), with research spanning topological machine learning, graph neural networks, and scientific visualization. My work develops theoretically grounded ML methods — leveraging topological priors, discrete Morse theory, persistence filtration, and hierarchical GNNs — to address core challenges in graph learning (oversmoothing, heterophily) and scientific image analysis across materials science, medicine, and computational biology. Published work includes topologically informed GNN architectures for segmentation and representation learning (topological deep learning and segmentation, ICIP/NeurIPS), generative modeling via conditional GANs and scientific computing and visualization in HPC environments (PAVE framework, IEEE/ACM), and deep learning for medical diagnostics (lymphoma classification, American Journal of Clinical Pathology). Current work centers on topological neural operators that extend operator learning to simplicial and cellular complexes via copresheaf transports, and on learned persistence-filtration based methods for improving graph neural networks under heterophily. I aim to build scalable, interpretable ML frameworks integrating topology, geometry, and generative AI to advance discovery in scientific visualization, medical diagnostics, and computational physics.

EDUCATION

<i>Ph.D. Computer Science</i>	2024	University of Utah School of Computing & Scientific Computing and Imaging Institute
<i>B.Sc. Physics Minor: Philosophy</i>	2013	University of Utah Department of Physics and Astronomy & College of Humanities
<i>B.Sc. Mathematics Minor: Comp. Sci.</i>	2013	University of Utah Department of Mathematics & School of Computing

DISSERTATION COMMITTEE

Valerio Pascucci (Chair), Christopher R. Johnson, Dave Pugmire, Bei Wang Phillips, Aditya Bhaskara

EXPERIENCE

2025– <i>present</i>	Postdoctoral Research – Topological Neural Operators, Neural operators extended to topological domains via Hodge-theoretic message passing on cochain complexes, with learnable copresheaf transports as discretization-invariant inductive biases. (NeurIPS 2026 submission; JMLR extended version in preparation). University of San Francisco supervisor: Mustafa Hajij
2017–24	Ph.D. Research Scientist University of Utah, Scientific Computing & Imaging Institute – Ph.D. mentor: Valerio Pascucci
2022	Visiting Research Scientist – GETO GNN: Topologically and Geometrically Constructed and Aware Graph Neural Networks Lawrence Livermore National Laboratory mentors: Mark Heimann and Jayaraman J. Thiagarajan

- 2021 Teaching Mentorship – Programming for Engineers.
Mentor: Valerio Pascucci
- 2020 Teaching Mentorship – Advanced Algorithms.
Mentor: Aditya Bhaskara
- 2019 Visiting Research Scientist – PAVE: An In Situ Framework for Scientific
Visualization and Machine Learning Coupling
Oak Ridge National Laboratory
Mentor: Mark Kim and David Pugmire
- 2018 High Dimensional Space Filling and Sampling Design with Improved
Spectral Sampling and Applications Monte-Carlo Integration.
Lawrence Livermore National Laboratory - Internship
Mentor: Jayaraman J. Thiagarajan
- 2016 Independent Study – Topological Distortion as a Quality Measure
for Dimension Reduction
University of Utah
mentor: Bei Wang
- 2010-12 Search for Quantum like Structuring in Keplerian Systems.
University of Utah - Award Funded Student Research
Mentor: Stephan LeBohec

AWARDS

- 2012 Fall Undergraduate Research Assistantship Award
Undergraduate Research Opportunities Program Assistantship.
- 2012 Spring Undergraduate Research Assistantship Award
Undergraduate Research Opportunities Program Assistantship.
- 2011 Scholar Athlete Award
Awarded by the Collegiate Water Polo Association.

OTHER EXPERIENCE

- 2023 Conference Presentation
IEEE International Conference on Image Processing
Graph Signal Processing and Machine Learning for
Interpretable and robust image processing II
- 2019 Conference Presentation
Super Computing Conference 2019, 5th International Workshop
on Data Analysis and Reduction for Big Scientific Data
(DRBSD-5)

PAPERS IN PREPARATION

- 2026 Topological Neural Operators
Leventhal, Samuel, Lennart Bastian, Tolga Birdal, and Mustafa Hajj. Conference version in
submission to NeurIPS 2026; extended version in preparation for JMLR.
- 2026 Homophily-based Filtration Learning For Graph Neural Networks
Leventhal, Samuel, Attila Gyulassy, Mark Heimann, and Valerio Pascucci. In preparation.

PUBLICATIONS

PEER REVIEWED JOURNAL PAPERS

- 2022 Exploring Classification of Topological Priors with Machine
Learning for Feature Extraction
Leventhal, Samuel, Attila Gyulassy, Mark Heimann, and Valerio Pascucci. "Exploring
Classification of Topological Priors with Machine Learning for Feature Extraction." IEEE
Transactions on Visualization and Computer Graphics (2023).

2020 Improving Augmented Human Intelligence to Distinguish Burkitt Lymphoma From Diffuse Large B-Cell Lymphoma Cases

Mohlman, Jeffrey S., Samuel D. Leventhal, Taft Hansen, Jessica Kohan, Valerio Pascucci, and Mohamed E. Salama. "Improving augmented human intelligence to distinguish Burkitt lymphoma from diffuse large B-cell lymphoma cases." *American journal of clinical pathology* 153, no. 6 (2020): 743-759.

2019 High-throughput feature extraction for measuring attributes of deforming open-cell foams.

Petruzza, Steve, Attila Gyulassy, Samuel Leventhal, John J. Baglino, Michael Czabaj, Ashley D. Spear, and Valerio Pascucci. "High-throughput feature extraction for measuring attributes of deforming open-cell foams." *IEEE Transactions on Visualization and Computer Graphics* 26, no. 1 (2019): 140-150.

2018 Application of a Convolutional Neural Network to Distinguish Burkitt Lymphoma From Diffuse Large B-Cell Lymphoma

Mohlman, Jeffrey, Samuel Leventhal, Aniketh Venkat, Attila Gyulassy, Valerio Pascucci, and Mohamed Salama. "Application of a Convolutional Neural Network to Distinguish Burkitt Lymphoma From Diffuse Large B-Cell Lymphoma." *American Journal of Clinical Pathology* 150, no. suppl.1 (2018): Page S119-S119.

2011 Entropy Is Not Our Anomie

Leventhal, Samuel. *the Sponge*, Volume 2, Issue 1. p5-p11. 2011

PEER REVIEWED CONFERENCE PAPERS

2023 Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks

Leventhal, Samuel, Attila Gyulassy, Valerio Pascucci, and Mark Heimann. "Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks." *International Conference on Image Processing*. 2023.

2013 A Search for Quantum Like Structuring in Keplerian Systems

Samuel Leventhal and Stephan LeBohec. "A Search For Quantum Like Structuring in Keplerian Systems." *University of Utah Undergraduate Research Abstracts Journal*, Vol. 13, Book 1. 2013

PEER REVIEWED WORKSHOP PAPERS

2022 Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks

Leventhal, Samuel, Attila Gyulassy, Valerio Pascucci, and Mark Heimann. "Modeling Hierarchical Topological Structure in Scientific Images with Graph Neural Networks." In *NeurIPS 2022 Workshop: New Frontiers in Graph Learning*. 2022.

2019 PAVE: An In Situ Framework for Scientific Visualization and Machine Learning Coupling

Leventhal, Samuel, Mark Kim, and David Pugmire. "Pave: An in situ framework for scientific visualization and machine learning coupling." In *2019 IEEE/ACM 5th International Workshop on Data Analysis and Reduction for Big Scientific Data (DRBSD-5)*, pp. 8-15. IEEE, 2019.